

A.R.T.I.C.U.L.A.T.E.

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NEED

Individuals with speech disabilities often struggle to be understood in their communities, impairing their independence. Most machine-read translators are text-to-speech, prohibiting illiterate individuals from use. Additionally, image-to-text devices are slow and limited in complexity.

TARGET AUDIENCE

Individuals who are unable to communicate easily, and cannot rely on text-based translation devices.

KEY FEATURES

- **Detects** speech and **processes** grammar
- Outputs a **corrected** version of given speech
- **Simple, accessible layout** with enhanced accessibility options

COMPETITORS



Voiceitt



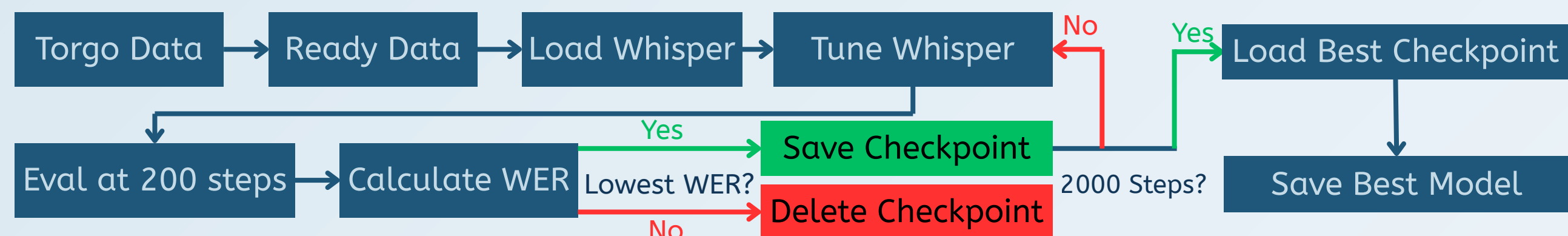
NOVA



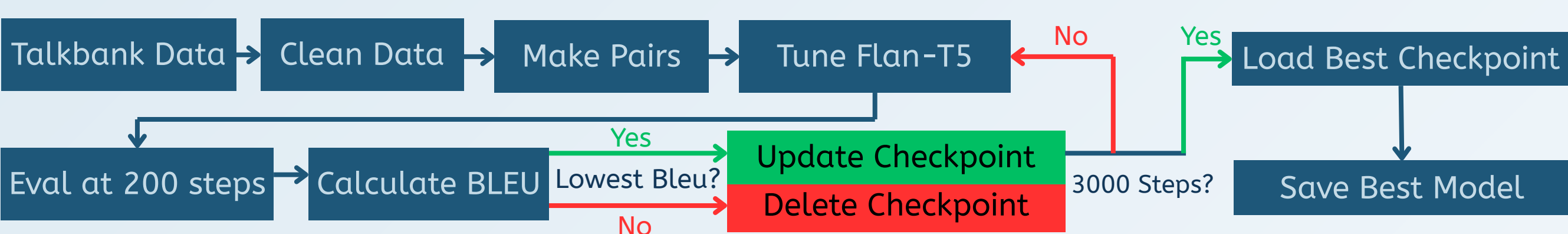
SmartBox

ARCHITECTURE

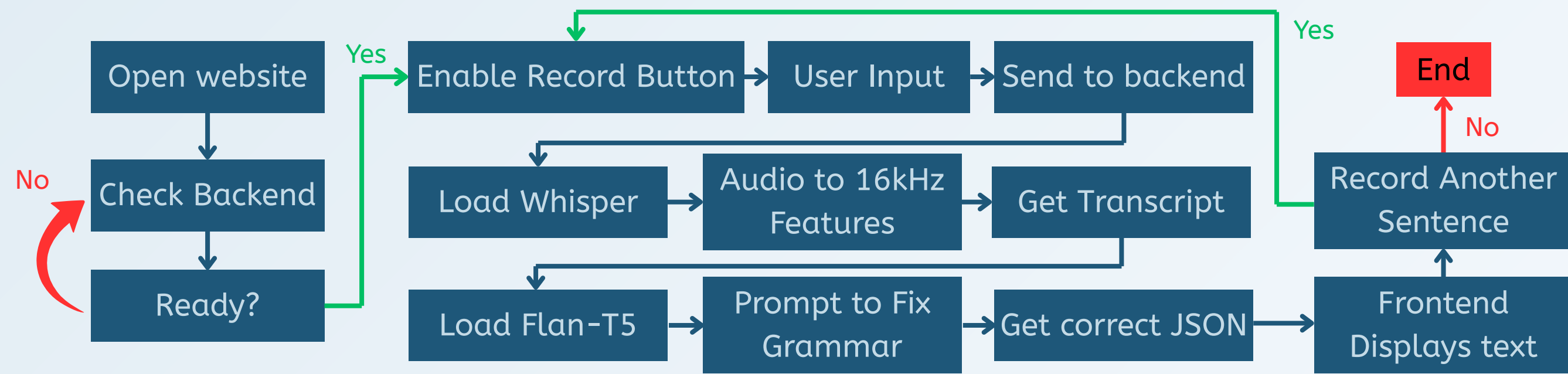
Whisper Model



Flan-T5 Model

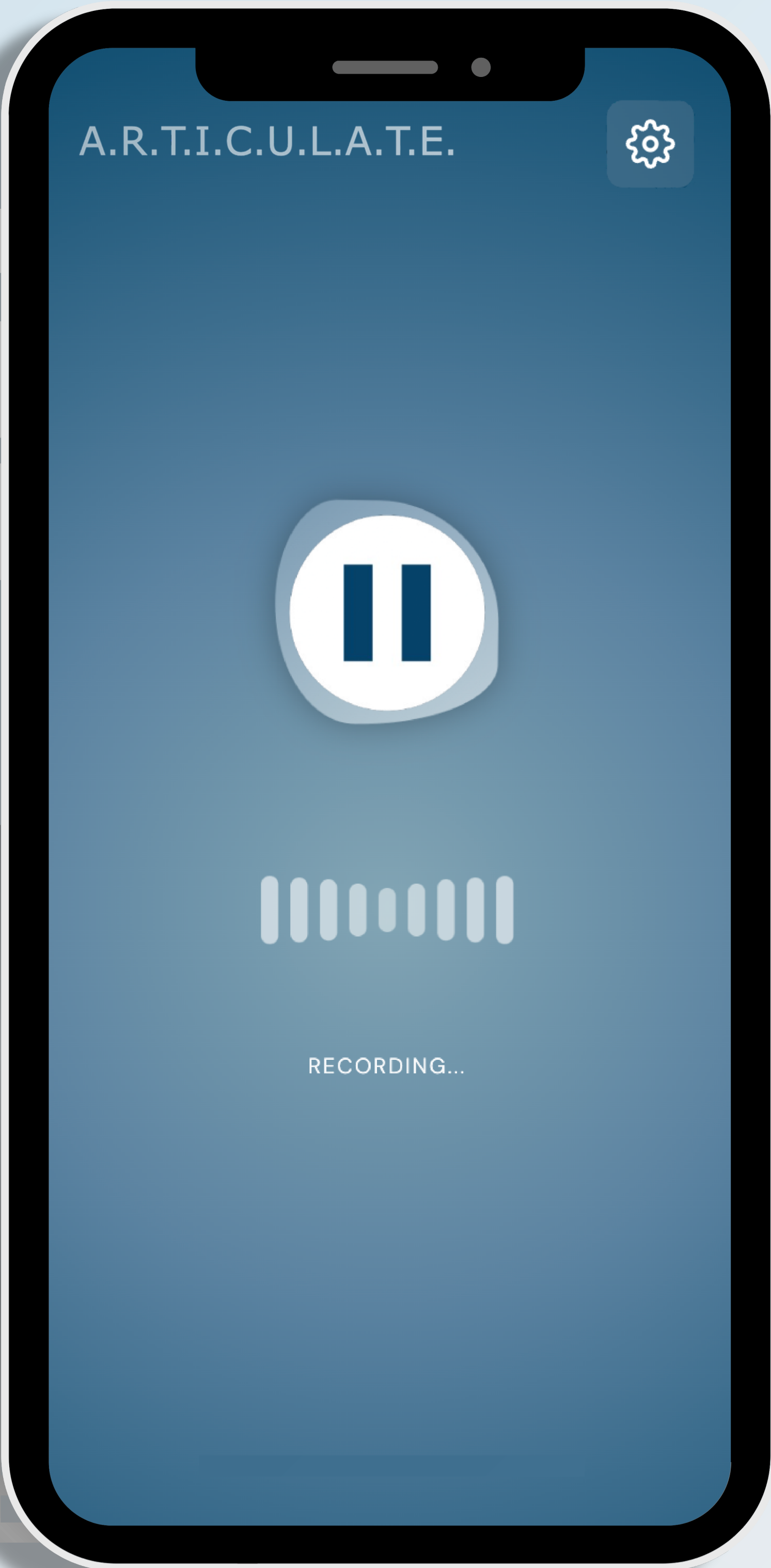


Website Flow



FUTURE WORK

- Train model with more speech disability-based audio
- Refine UI with application-oriented software
- Publish ARTICULATE on the App Store
- Develop a “user-profile” and program to track speech progress



Programs and Resources Used:

